……[2]

5 **(a)** This question is about an americium-241 radionuclide source used in a smoke detector.

A smoke detector works by detecting a decrease in the arrival of alpha particles from the americium when they are absorbed by the smoke. It is advised that the smoke detector be replaced after five years.

Here are some data about the source:
activity of source = 3.3×10^4 Bq
decay constant = 4.8×10^{-11} s^{-1}

(i) Radioactive decay is *random* in nature. State what is meant by the word *random* in this context.

……

……[1]

(ii) Use the equation $\Delta N = -\lambda N \Delta t$ with N taken as the original number of nuclei to show that about 5×10^{12} nuclei decay in the five years of use.

[2]

(iii) Explain why it is reasonable to use the equation $\Delta N = -\lambda N \Delta t$ (as above) to estimate the number of nuclei decaying over a five year period, but not over a few hundred years.

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……[2]

(iv) A decrease in activity of the sample is not likely to be the reason that the smoke detector should be replaced after five years. Suggest a more likely reason that the detector might need replacing.

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……[2]

(b) Strontium-90 is a radioactive waste product of nuclear power stations. Strontium-90 emits beta particles of energy up to 0.54 MeV. Should any of this isotope get into our food, it would be quickly absorbed into bones and teeth.
Explain why these beta particles are more of a risk to humans than gamma photons of the same energy.

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……[2]

6 **(a)** A dust particle is suspended in the air between two parallel plates with the direction